

Module 2 — Synthesis Quiz

[Question bank](#) · [After exercise lab](#) · [open notes](#) · [two attempts](#)

Module 2, Synthesis quiz

Readings: [Read 0](#), [Read 1](#), [Read 2](#)

Labs: [Demo 1 — tools](#), [Demo 2 — worked](#), [Exercise](#)

Canvas: Synthesis quiz (Module 2) · two attempts, highest score counts

Full integration for [Project 2](#): Demo 1 R tools (`binom.test`, `prop.test`, normal overlays, `ggplot`, `t.test`, FPC/survey), Demo 2 Read 2 workbook + CSV workflow, exercise lab parallel stories, reading R output (Section C), Type I/II errors, and Read 1–2 conditions & interpretation.

Instructions

- Open notes ([methods map](#), [R reference](#)).
 - Use the CI interpretation template from Read 1: “*I am 95% confident that...*”
 - For R items, match [Demo 1](#), [Demo 2](#), and your [Project 2 templates](#).
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Section A — Interpret confidence intervals

Use: “*I am 95% confident that [parameter] in the entire population is between [L] and [U]. In this context, confidence means that if data from many samples were collected the same way, then 95% of the intervals computed from these samples would contain the true parameter.*”

A1. First-generation proportion

95% CI for proportion of undergraduates who are first-generation: [0.22, 0.30]. Interpret in context.

A2. Website minutes mean

95% CI for mean minutes/day on course websites: [41, 55] minutes. Interpret.

A3. Schools older than 50 years

95% CI with FPC for proportion of schools older than 50 years: [0.14, 0.27]. Interpret.

A4. Spot incorrect interpretation

Which is the best interpretation of a 95% CI?

- A) 95% probability the parameter is in this interval.
 - B) Repeated sampling: ~95% of such intervals contain the parameter.
 - C) 95% of individuals fall between the endpoints.
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Section B — Demo 1 tools (R output & choices)

See [Demo lab 1 — tools](#). Read printed output — do not recompute by hand unless asked.

B1. binom vs prop

$n = 16$, $x = 3$ deaths (mortality proportion). Which function for a 95% CI and why?

B2. prop.test output

`prop.test(x=332, n=400, correct=FALSE)` (seatbelt use). 1. Parameter? 2. Approximate 95% CI? 3. Large-count OK?

B3. t.test output

`t.test(c(8,10,12,12,14,16))` for library hours. Parameter? FPC needed if $N = 120$ and SRS without replacement?

B4. Normal tools analogy

For numerical outcomes, which pair is the normal analogue of `dbinom` / `pbinom` for counts?

B5. Dot plot vs histogram

You have $n = 6$ library hour values. Which ggplot layer shows each duplicate clearly?

B6. survey FPC

In `svydesign(..., fpc = ~N)`, what does the `fpc` argument tell R?

B7. prop.test() decision

When using `prop.test(x, n, p = p0)` for a hypothesis test, which output do you compare to α ?

Section C — Read R output, check conditions & graphs

Read `prop.test()` and `t.test()` output like [Demo 1](#). Check the [conditions checklist](#) before trusting an interval. Image items use the same figures as Canvas.

C1. prop.test — sample proportion

From this output, what is the sample proportion $\hat{p}$?

```
prop.test(x = 332, n = 400, correct = FALSE)
...
sample estimates:
  prop with continuity correction  0.83
```

A) 0.332

B) 0.83

C) 0.17

D) 1.96

C2. `prop.test` — 95% CI endpoints

Using the same seatbelt study (`x = 332, n = 400`), R prints:

```
95 percent confidence interval:
 0.7879343 0.8720657
```

Which is the best approximate 95% CI for the population proportion?

- A) $[0.79, 0.87]$
- B) $[0.33, 0.40]$
- C) $[0.79, 0.83]$
- D) $[332, 400]$

C3. `prop.test` — p-value conclusion

A one-sided test uses `prop.test(x = 62, n = 100, p = 0.5, alternative = "greater")` and reports p-value $< 2.2\text{e-}16$. At $\alpha = 0.05$, what is the best conclusion about $H_0 : p = 0.5$ vs $H_a : p > 0.5$?

- A) Reject H_0 — strong evidence the population proportion exceeds 0.5
- B) Fail to reject H_0 — not enough evidence
- C) Accept H_0 as proven true
- D) The p-value is the sample proportion

C4. Wald large-count check

For `prop.test` / Wald on $n = 400$, $x = 332$, do large-count conditions hold?

- A) Yes — $n\hat{p}$ and $n(1 - \hat{p})$ are both ≥ 10
- B) No — need $n > 400$
- C) No — only successes matter
- D) Yes — only when FPC is applied

C5. `t.test` — sample mean

Library hours for $n = 6$ campuses:

```
t.test(c(8, 10, 12, 12, 14, 16))
...
mean of x          12
```

What is the single best estimate of the population mean from these data?

- A) $\mu = 12$ (known parameter)
- B) $\bar{x} = 12$ (sample statistic)
- C) Margin of error = 12
- D) p-value = 12

C6. `t.test` — interpret CI

For the same library-hour sample, R reports:

```
95 percent confidence interval:
 9.483032 14.516968
```

Best one-sentence interpretation?

- A) I am 95% confident the true population mean weekly open hours is between 9.48 and 14.52 hours.
- B) There is a 95% probability μ is in this interval.
- C) 95% of libraries have hours in this range.
- D) We proved $\mu = 12$ exactly.

C7. `t.test` — when and why

SRS of $n = 6$ library hours from finite N ; σ unknown. Why is `t.test()` appropriate?

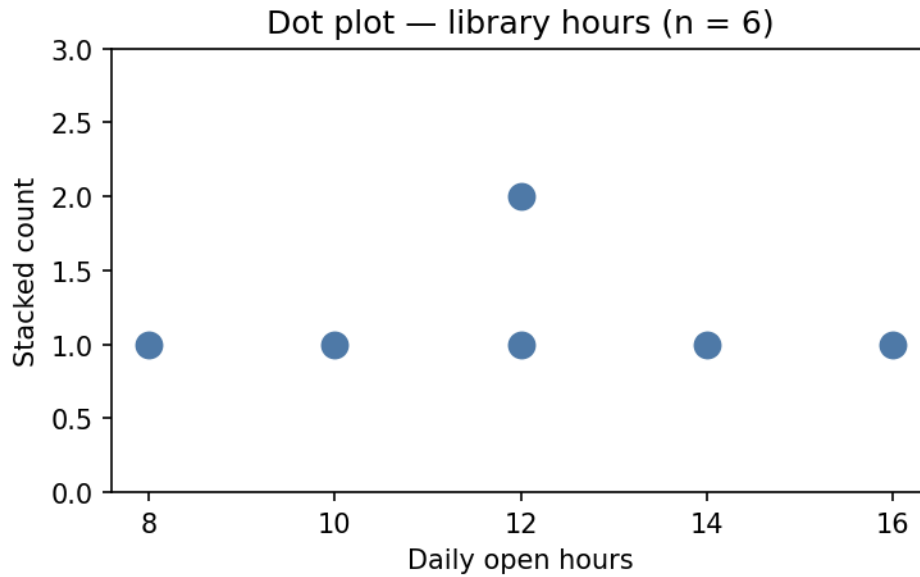
- A) s is estimated from the sample; t^* adjusts for small n when σ is unknown
- B) Proportions always need `binom.test()`
- C) FPC replaces the need for any conditions
- D) Mean CIs never require a sampling design check

C8. `binom.test` vs `prop.test`

Hospital mortality: $n = 16$ patients, $x = 3$ deaths. Best 95% CI function?

- A) `binom.test()` — Wald large-count conditions fail
- B) `prop.test()` only — always Wald
- C) `t.test()`
- D) No CI is possible

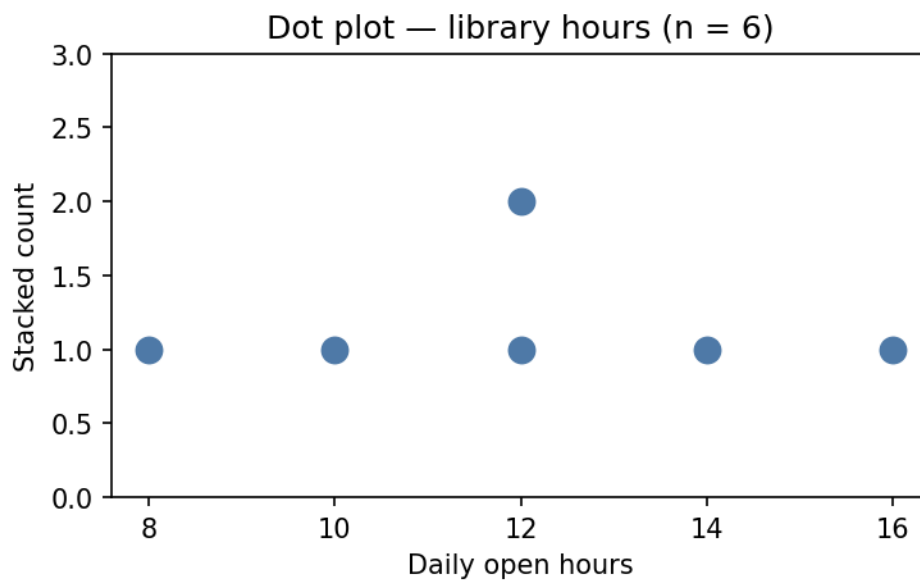
C9. Dot plot — sample mean



Six sampled libraries gave daily open hours (hours) in the dot plot. What is $\bar{x}$?

- A) 8
- B) 10
- C) 12
- D) 14

C10. Dot plot — ggplot layer



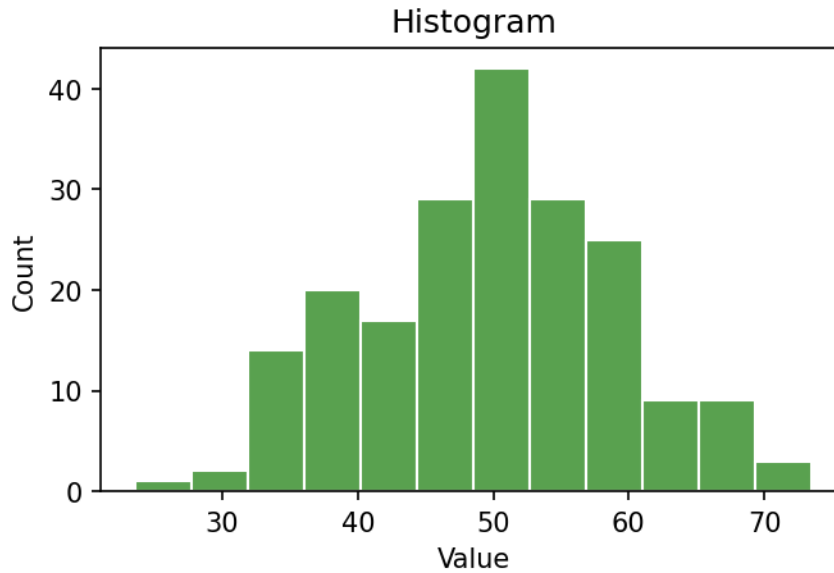
Which ggplot2 layer creates stacked dots at each observed value (good for small n)?

- A) `geom_dotplot()`
- B) `geom_histogram()`

C) `geom_bar()`

D) `geom_smooth()`

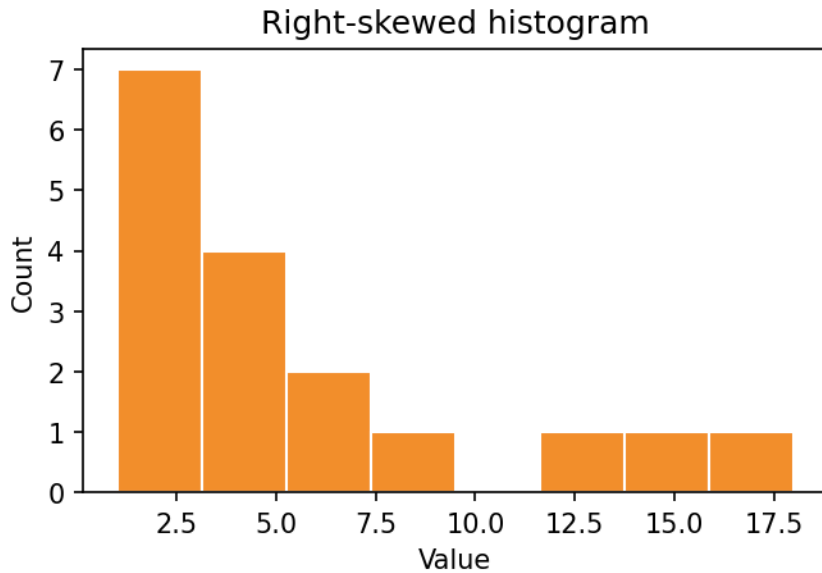
C11. Histogram — what it shows



What does this graph primarily display?

- A) The distribution of one numerical variable
- B) A 95% CI for a proportion
- C) A two-way table of proportions
- D) A hypothesis-test p-value

C12. Histogram — skew and center



This histogram is right-skewed. For a right-skewed distribution, which is typically larger?

- A) Mean > median
 - B) Median > mean
 - C) Mean = median always
 - D) Cannot compare from a histogram
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Section D — Type I/II errors in context

D1. Lucky coin

Test if coin favors heads ($p = P(\text{heads})$). State H_0 , H_a , Type I error, Type II error in context.

D2. Lucky die

Test if die favors 6. State H_0 , H_a , Type I and Type II errors in context.

D3. Medical treatment

Test if treatment effective in more than 10% of cases. State hypotheses and both error types.

D4. Left-handed students

Test if school left-handed rate exceeds 10%. State hypotheses and both error types.

D5. Defective supplier

Test if defect rate below 15%. State hypotheses and both error types.

Section E — Decisions & possible errors

Use $\alpha = 0.05$ for each problem.

E1. 62 heads in 100

$H_0 : p = 0.5$ vs $H_a : p > 0.5$; 62 heads; p-value = 0.0096.

Reject? Type I possible? Type II possible? State Type I in context.

E2. 4 left-handed in 25

$H_0 : p = 0.10$ vs $H_a : p > 0.10$; 4/25 left-handed; p-value = 0.21.

Reject? Type I possible? Type II possible? State Type II in context.

E3. 3 defective in 40

$H_0 : p = 0.15$ vs $H_a : p < 0.15$; 3/40 defective; p-value = 0.034.

Reject? Type I possible? Type II possible? State Type I in context.

Section F — Reducing errors

F1. Reduce Type I

Which helps reduce Type I error?

A) Larger α B) Smaller α C) Increase n D) Decrease n

F2. Reduce Type II (fixed α)

With fixed α , what increases power / reduces Type II error?

A) Smaller α B) Larger α C) Increase n D) Decrease n

Section G — Demo 2 worked examples (Read 2 workbook)

Match each **Demo 2** / Read 2 example to the correct interval family, conditions, and FPC decision.

G1. Example 1 — seatbelts

SRS $n = 400$, $x = 332$ seatbelt users; statewide population.

Parameter? Method? FPC?

G2. Example 2 — St. Joe's mortality

$n = 16$, $x = 3$ deaths.

Which method? Why not Wald?

G3. Example 3 — plant-based entrée

$N = 80$, $n = 20$, $x = 6$ yes.

Parameter? FPC? Borderline Wald note?

G4. Example 4 — library hours

$N = 120$, $n = 6$ hours: 8,10,12,12,14,16.

Parameter? FPC? t vs z ?

G5. Example 5 — commute times

$n = 40$, $\bar{x} = 28.5$ min, $s = 8.2$; large metro.

Parameter? FPC? CLT role?

Section H — Exercise lab (new stories)

Parallel to **exercise lab** — application and interpretation.

H1. Flu vaccine proportion

$n = 350$, $x = 266$ vaccinated; large state population.

Method? One-sentence CI interpretation template?

H2. Drug side effect

$n = 14$, $x = 2$ with side effect.

Check conditions. Preferred CI function?

H3. Dorm study lounge

$N = 95$, $n = 22$, $x = 8$ with lounge.

FPC factor formula? Why multiply SE?

H4. Espresso temperature mean

$N = 60$ shops, $n = 12$ temperatures ($^{\circ}\text{F}$). Which plot first? Which CI family?

H5. Daily steps mean

$n = 35$ step counts; large app cohort.

FPC? Conditions checklist (name two)?

Section I — Read 1–2 application & conditions

I1. Test vs CI (proportion)

You want a 95% CI for p only (no p_0). Do you pass `p = to prop.test`?

I2. Wald by hand

$\hat{p} = 0.83$, $n = 400$. Compute approximate SE and MOE (multiplier 2).

I3. Plus Four when

When is Plus Four a reasonable alternative to exact binomial for a proportion CI?

I4. Population vs sample

$\bar{x} = 12$ library hours from $n = 6$ libraries. Is 12 a parameter or statistic?

I5. FPC threshold

When should you consider applying the FPC in Project 2?

I6. Wrong CI language

Student writes: ‘There is a 95% chance the true mean is in $[25.9, 31.1]$.’ What is wrong?
